P2 Chapter 11: Integration

Finding Areas

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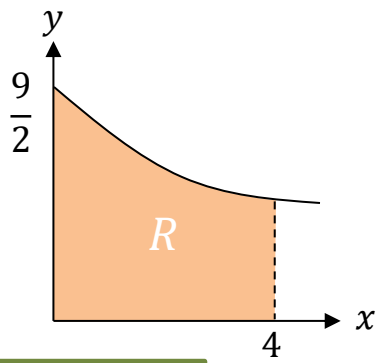
You're already familiar with the idea that definite integration gives you the (signed) area bound between the curve and the x-axis.
Given your expanded integration skills, you can now find the area under a greater variety of curves.

[Textbook] The diagram shows part of the curve $y = \frac{9}{\sqrt{4+3x}}$
The region R is bounded by the curve, the x-axis and the lines $x = 0$ and $x = 4$, as shown in the diagram. Use integration to find the area of R.



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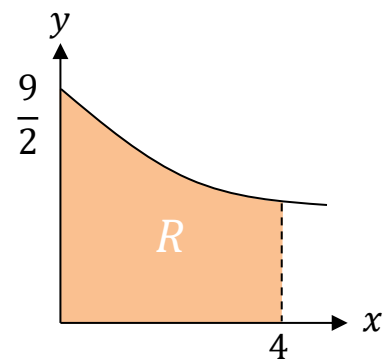
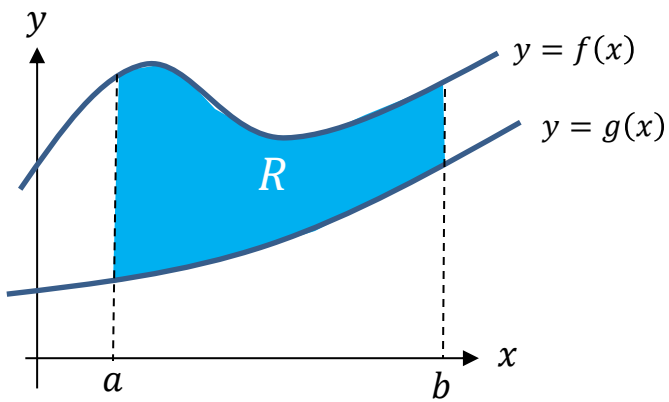


$$\boldsymbol{Area = \int_0^4 \frac{9}{\sqrt{4+3x}}\,dx}$$
$$\boldsymbol{= 9\int_0^4 (4+3x)^{-\frac{1}{2}}\,dx}$$
$$\boldsymbol{= 6\left[(4+3x)^{\frac{1}{2}}\right]_0^4}$$
$$\boldsymbol{= \cdots = 12}$$

Recall that when we integrate $(a+bx)^n$, we divide both by the increased power $n+1$, but also by b.

Area between Two Curves



Tip: Ensure you have top curve minus bottom curve.

(This was presented in my Year 1 slides as an 'alternative method')

The areas under the two curves are $\int_a^b f(x)\,dx$ and $\int_a^b g(x)\,dx$. It therefore follows the area between them (provided the curves don't overlap) is:

$$R = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$$

$$= \int_a^b \big(f(x) - g(x)\big)\,dx$$

[Textbook] The diagram shows part of the curves $y = \sin 2x$ and $y = \sin x \cos^2 x$ where $0 \le x \le \frac{\pi}{2}$. The region R is bounded by the two curves. Use integration to find the area of R.

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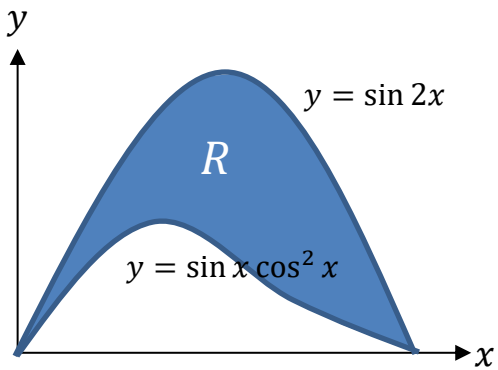

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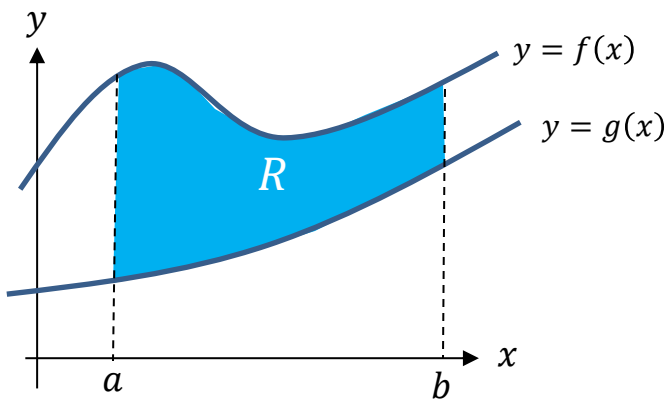

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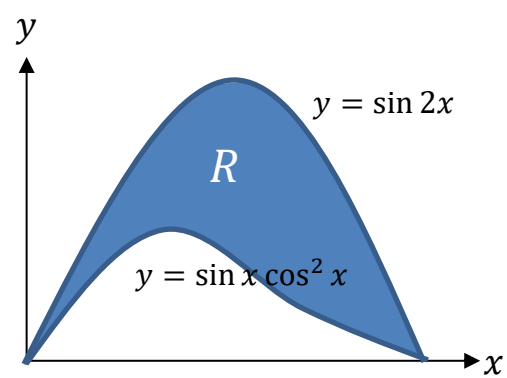

$$\boldsymbol{Area = \int_0^{\frac{\pi}{2}} \sin 2x - \sin x \cos^2 x \;\; dx}$$

$$\boldsymbol{= \left[-\frac{1}{2}\cos 2x + \frac{1}{3}\cos^3 x\right]_0^{\frac{\pi}{2}} = \cdots = \frac{2}{3}}$$

Test Your Understanding

Edexcel C4 Jan 2009 Q2

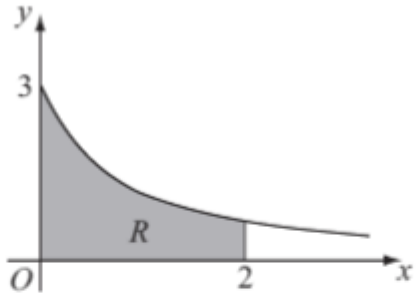

Figure 1

Figure 1 shows part of the curve $y = \dfrac{3}{\sqrt{(1+4x)}}$. The region R is bounded by the curve, the x-axis, and the lines $x = 0$ and $x = 2$, as shown shaded in Figure 1.

(a) Use integration to find the area of R. **(4)**

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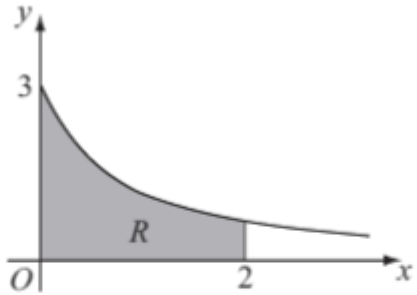


Figure 1

Figure 1 shows part of the curve $y = \dfrac{3}{\sqrt{(1+4x)}}$. The region R is bounded by the curve, the x-axis, and the lines $x = 0$ and $x = 2$, as shown shaded in Figure 1.

(a) Use integration to find the area of R. **(4)**

$$\text{Area}(R) = \int_0^2 \frac{3}{\sqrt{(1+4x)}}\,dx = \int_0^2 3(1+4x)^{-\frac{1}{2}}\,dx$$

Integrating $3(1+4x)^{-\frac{1}{2}}$ to give $\pm k(1+4x)^{\frac{1}{2}}$. — M1

$$= \left[\frac{3(1+4x)^{\frac{1}{2}}}{\frac{1}{2}.4}\right]_0^2$$

Correct integration. Ignore limits. — A1

$$= \left[\tfrac{3}{2}(1+4x)^{\frac{1}{2}}\right]_0^2$$

$$= \left(\tfrac{3}{2}\sqrt{9}\right) - \left(\tfrac{3}{2}(1)\right)$$

Substitutes limits of 2 and 0 into a changed function and subtracts the correct way round. — M1

$$= \tfrac{9}{2} - \tfrac{3}{2} = \underline{3}\,(\text{units})^2$$

$\underline{3}$ — A1

Exercise 11.8

Pearson Pure Mathematics Year 2
Pages 96-98

Homework Exercise

1 Find the area of the finite region R bounded by the curve with equation $y = \mathrm{f}(x)$, the x-axis and the lines $x = a$ and $x = b$.

a $\mathrm{f}(x) = \dfrac{2}{1+x}$; $a = 0$, $b = 1$

b $\mathrm{f}(x) = \sec x$; $a = 0$, $b = \dfrac{\pi}{3}$

c $\mathrm{f}(x) = \ln x$; $a = 1$, $b = 2$

d $\mathrm{f}(x) = \sec x \tan x$; $a = 0$, $b = \dfrac{\pi}{4}$

e $\mathrm{f}(x) = x\sqrt{4 - x^2}$; $a = 0$, $b = 2$

2 Find the exact area of the finite region bounded by the curve $y = \mathrm{f}(x)$, the x-axis and the lines $x = a$ and $x = b$ where:

a $\mathrm{f}(x) = \dfrac{4x - 1}{(x+2)(2x+1)}$; $a = 0$, $b = 2$

b $\mathrm{f}(x) = \dfrac{x}{(x+1)^2}$; $a = 0$, $b = 2$

c $\mathrm{f}(x) = x \sin x$; $a = 0$, $b = \dfrac{\pi}{2}$

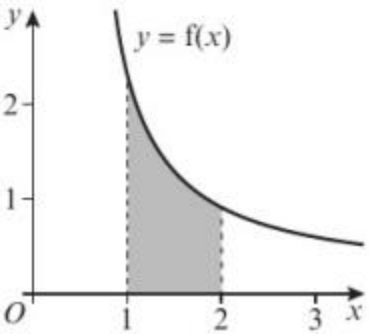
d $\mathrm{f}(x) = \cos x\sqrt{2\sin x + 1}$; $a = 0$, $b = \dfrac{\pi}{6}$

e $\mathrm{f}(x) = xe^{-x}$; $a = 0$, $b = \ln 2$

3 The diagram shows a sketch of the curve with equation, $y = \mathrm{f}(x)$,

where $\mathrm{f}(x) = \dfrac{4x + 3}{(x+2)(2x-1)}$, $x > \frac{1}{2}$

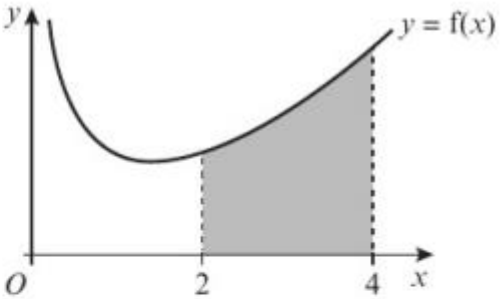
Find the area of the shaded region bounded by the curve, the x-axis and the lines $x = 1$ and $x = 2$. **(7 marks)**



Homework Exercise

4 The diagram shows a sketch of the curve with equation $y = f(x)$, where $f(x) = e^{0.5x} + \frac{1}{x}$, $x > 0$.

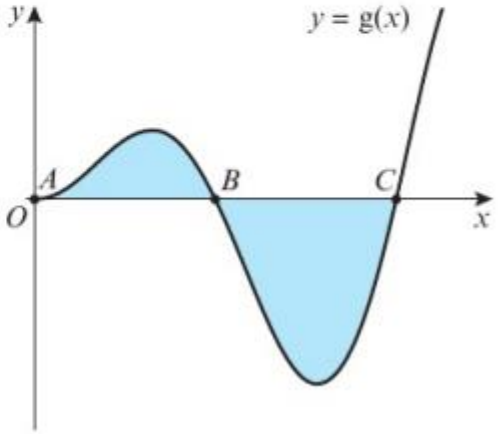
Find the area of the shaded region bounded by the curve, the x-axis and the lines $x = 2$ and $x = 4$. **(7 marks)**



5 The diagram shows a sketch of the curve with equation $y = g(x)$, where $g(x) = x \sin x$.

a Write down the coordinates of points A, B and C.

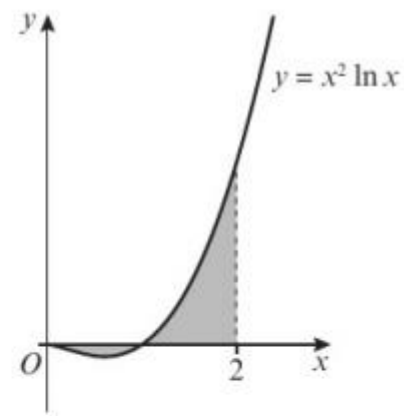
b Find the area of the shaded region.



Watch out Find the area of each region separately and then add the answers. Remember areas cannot be negative, so take the absolute value of any negative area.

Homework Exercise

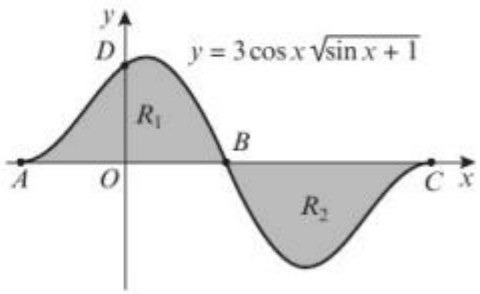
6 The diagram shows a sketch of the curve with equation $y = x^2 \ln x$. The shaded region is bounded by the curve, the x-axis and the line $x = 2$.



a Use integration by parts to find $\int x^2 \ln x \, dx$. **(3 marks)**

b Hence find the exact area of the shaded region, giving your answer in the form $\frac{2}{3}(a \ln 2 + b)$, where a and b are integers. **(5 marks)**

7 The diagram shows a sketch of the curve with equation $y = 3\cos x\sqrt{\sin x + 1}$.



a Find the coordinates of the points A, B, C and D. **(3 marks)**

b Use a suitable substitution to find

$\int 3\cos x\sqrt{\sin x + 1} \, dx$ **(5 marks)**

c Show that the regions R_1 and R_2 have the same area, and find the exact value of this area in the form $\sqrt{a}$, where a is a positive integer to be found. **(3 marks)**

8 $f(x) = x^2$ and $g(x) = 3x - x^2$

a On the same axes, sketch the graphs of $y = f(x)$ and $y = g(x)$, and find the coordinates of any points of intersection of the two curves.

b Find the area of the finite region bounded by the two curves.

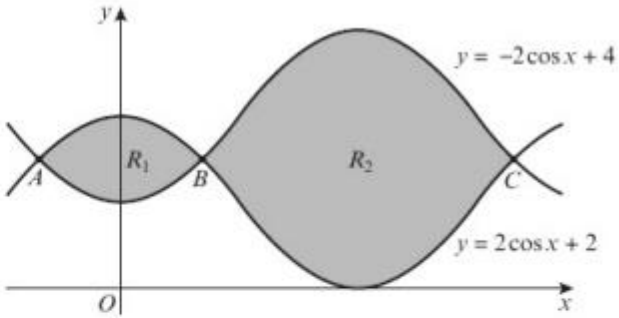
Homework Exercise

9 The diagram shows a sketch of part of the curves with equations $y = 2\cos x + 2$ and $y = -2\cos x + 4$.

a Find the coordinates of the points A, B and C. **(2 marks)**

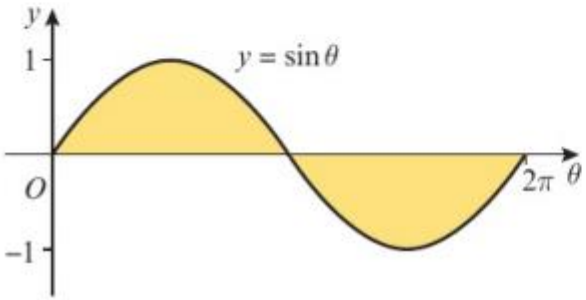
b Find the area of region R_1 in the form $a\sqrt{3} + \frac{b\pi}{c}$, where a, b and c are integers to be found. **(4 marks)**

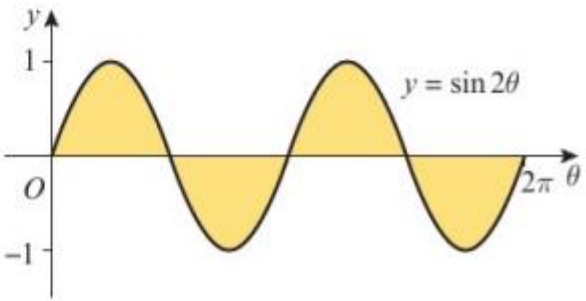
c Show that the ratio of $R_2 : R_1$ can be expressed as $(3\sqrt{3} + 2\pi) : (3\sqrt{3} - \pi)$. **(5 marks)**



10 The diagrams show the curves $y = \sin\theta$, $0 \leqslant \theta \leqslant 2\pi$ and $y = \sin 2\theta$, $0 \leqslant \theta \leqslant 2\pi$.

By choosing suitable limits, show that the total shaded area in the first diagram is equal to the total shaded area in the second diagram, and state the exact value of this shaded area.





Homework Exercise

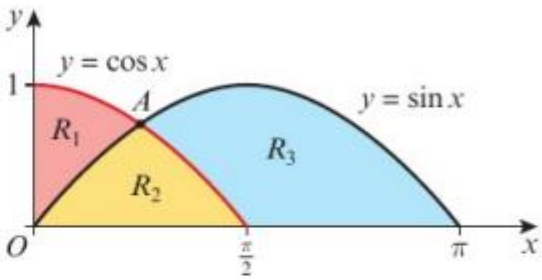
11 The diagram shows parts of the graphs of $y = \sin x$ and $y = \cos x$.

a Find the coordinates of point A.

b Find the areas of:

i R_1 **ii** R_2 **iii** R_3

c Show that the ratio of areas $R_1 : R_2$ can be written as $\sqrt{2} : 2$.

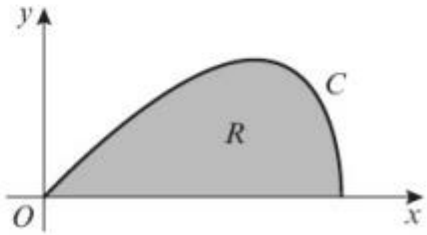


12 The curve C has parametric equations $x = t^3$, $y = t^2$, $t \geqslant 0$. Show that the exact area of the region bounded by the curve, the x-axis and the lines $x = 0$ and $x = 4$ is $k\sqrt[3]{2}$, where k is a rational constant to be found.

13 The curve C has parametric equations

$$x = \sin t,\ y = \sin 2t,\ 0 \leqslant t \leqslant \frac{\pi}{2}$$

The finite region R is bounded by the curve and the x-axis. Find the exact area of R. **(6 marks)**



Homework Exercise

14 This graph shows part of the curve C with parametric equations $x = (t + 1)^2$, $y = \frac{1}{2}t^3 + 3$, $t \geqslant -1$

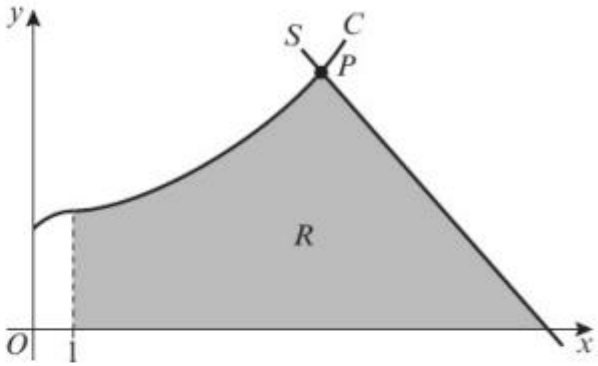
P is the point on the curve where $t = 2$.

The line S is the normal to C at P.

a Find an equation of S. **(5 marks)**

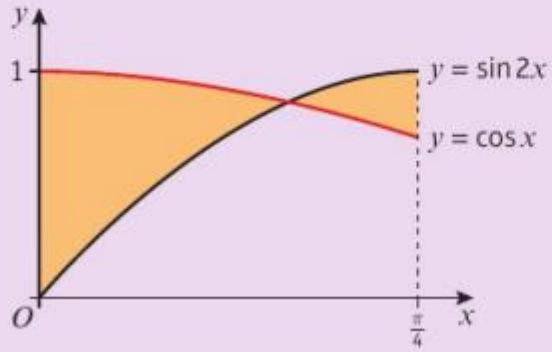
The shaded region R is bounded by C, S, the x-axis and the line with equation $x = 1$.

b Using integration, find the area of R. **(5 marks)**



Challenge

The diagram shows the curves $y = \sin 2x$ and $y = \cos x$, $0 \leqslant x \leqslant \frac{\pi}{4}$



Find the exact value of the total shaded area on the diagram.

Homework Answers

1 **a** $2\ln 2$ **b** $\ln(2+\sqrt{3})$ **c** $2\ln 2 - 1$
d $\sqrt{2} - 1$ **e** $\frac{8}{3}$

2 **a** $\ln\frac{8}{5}$ **b** $\ln 3 - \frac{2}{3}$ **c** 1
d $\dfrac{2\sqrt{2}-1}{3}$ **e** $\frac{1}{2}(1 - \ln 2)$

3 $\ln 4$ **4** $2e^2 - 2e + \ln 2$

5 **a** $A(0, 0)$, $B(\pi, 0)$ and $C(2\pi, 0)$

b $\text{Area} = \int_0^{\pi} x\sin x\,dx - \int_{\pi}^{2\pi} x\sin x\,dx$

$= [-x\cos x + \sin x]_0^{\pi} - [-x\cos x + \sin x]_{\pi}^{2\pi}$

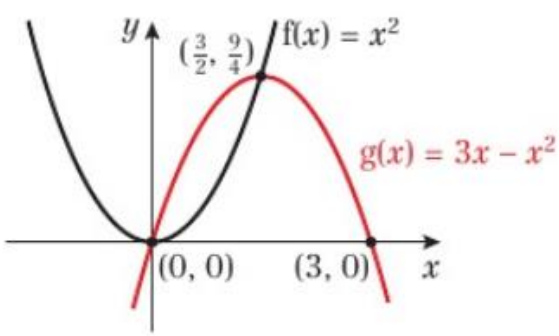
$= \pi + 3\pi = 4\pi$

6 **a** $\frac{1}{3}x^3\ln x - \frac{1}{9}x^3 + c$ **b** $\frac{2}{3}(4\ln 2 - 1)$

7 **a** $A\left(-\frac{\pi}{2}, 0\right)$, $B\left(\frac{\pi}{2}, 0\right)$, $C\left(\frac{3\pi}{2}, 0\right)$ and $D(0, 3)$

b $2(\sin x + 1)^{\frac{3}{2}} + c$ **c** $a = 32$

8 **a**



b $\frac{9}{8}$

9 **a** $A\left(-\frac{\pi}{3}, 3\right)$, $B\left(\frac{\pi}{3}, 3\right)$ and $C\left(\frac{5\pi}{3}, 3\right)$

b $a = 4, b = -4, c = 3$ (or $a = 4, b = 4, c = -3$)

c $R_2 = \int_{\frac{\pi}{3}}^{\frac{5\pi}{3}} (-2\cos x + 4)\,dx - \int_{\frac{\pi}{3}}^{\frac{5\pi}{3}} (2\cos x + 2)\,dx$

$= \int_{\frac{\pi}{3}}^{\frac{5\pi}{3}} (-4\cos x + 2)\,dx = \left[-4\sin x + 2x\right]_{\frac{\pi}{3}}^{\frac{5\pi}{3}}$

$= \left(2\sqrt{3} + \frac{10\pi}{3}\right) - \left(-2\sqrt{3} + \frac{2\pi}{3}\right) = 4\sqrt{3} + \frac{8\pi}{3}$

$R_2 : R_1 \Rightarrow 4\sqrt{3} + \frac{8\pi}{3} : 4\sqrt{3} - \frac{4\pi}{3} \Rightarrow 3\sqrt{3} + 2\pi : 3\sqrt{3} - \pi$

10 $y = \sin\theta : \text{Area} = \int_0^{\pi} 2\sin\theta\,d\theta = [-2\cos\theta]_0^{\pi} = (2) - (-2) = 4$

$y = \sin 2\theta : \text{Area} = \int_0^{\frac{\pi}{2}} 4\sin 2\theta\,d\theta = \left[-2\cos 2\theta\right]_0^{\frac{\pi}{2}}$

$= (2) - (-2) = 4$

11 **a** $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$

b **i** $\sqrt{2} - 1$ **ii** $2 - \sqrt{2}$ **iii** $\sqrt{2}$

c $R_1 : R_2 \Rightarrow \sqrt{2} - 1 : 2 - \sqrt{2}$

$\Rightarrow (\sqrt{2} - 1)(2 + \sqrt{2}) : (2 - \sqrt{2})(2 + \sqrt{2}) \Rightarrow \sqrt{2} : 2$

Homework Answers

12 Area $= \int y \frac{dx}{dt}\, dt = \int_0^{\sqrt[3]{4}} t^2(3t^2)\, dt = \frac{3}{5}(\sqrt[3]{4})^5 = \frac{3}{5}\, 2^{\frac{10}{3}}$

$= \frac{3}{5}(2^3)(2^{\frac{1}{3}}) = \frac{24}{5}\sqrt[3]{2}$

13 $\frac{2}{3}$

14 **a** $x + y = 16$ **b** 58.9

Challenge

Area of region $R = \dfrac{2 - \sqrt{2}}{2}$.